INCH-POUND

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16 June 1989
SUPERSEDING
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MILITARY SPECIFICATION

TRUCK, LIFT, WHEEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a manually operated mechanical or manually operated hydraulic wheel lift truck for removing, replacing, and conveying automotive-type wheel and tire assemblies. The wheel lift truck shall have a load capacity of not less than 1500 pounds (lbs) and a safety proof load of 1875 lbs.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents are listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-60 - Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-H-6083 - Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-F-17111 - Fluid, Power Transmission.

STANDARDS

FEDERAL

FED-STD-H25 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods.
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-509?.)

2.1.2 Other Government documents, drawings, and publicationns. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402-0001.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subject to first article inspection (see 6.4) in accordance with 4.3. The approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

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3.2 Design. The wheel lift shall be new and of the manufacturer's current commercial design. It shall be used to facilitate the removal, installation, and conveyance of single and dual wheel and tire assemblies of automotive vehicles. The lift shall be a manually operated mechanical or manually operated hydraulic mechanism. All parts subject to wear shall be accessible for replacement, adjustment, and repair without modification to the wheel lift truck.

3.3 Materials. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practices, shall be suitable for the intended purpose in the design and fabrication of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or longevity of the individual components or the end item equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Raw materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4 Fabrication. The end item of equipment shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the end item of equipment, to its maximum rating and capacity, without permanent distortion.

3.4.1 Castings and forgings. Finished castings and forgings shall be free of fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, or other defects. No processes such as welding, peening, plugging, soldering, or pasting shall be used for repairing, reworking, or improving cosmetic appearance.

3.4.2 Welding, brazing, or soldering. All welding, brazing, or soldering shall be free of defects and of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. Defects shall include but are not limited to such things as bubbles, cracks, gouges, holes, insufficient materials used. All flux or other such materials used during these operations shall be thoroughly removed (cleaned), from the part(s) immediately upon completion of the operation. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Fastening devices and methods shall be chosen to serve the need while providing necessary adjustability for service, maintenance of repair. Fasteners shall be installed to prevent loss of tightness and shall not loosen in service.

3.4.4 Threads. All machined threads shall conform to FED-STD-H28 and the applicable detailed standards referenced therein.

3.4.5 Lubrication. Means shall be provided for lubrication except permanently lubricated sealed-for-life mechanisms.

3.4.6 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide easy access to guarded parts. The wheel lift truck shall comply with the general safety and

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health requirements promulgated under 29CFR 1910 that are applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified in the contract. (see 6.2.1 and 6.5).

3.4.6.1 Over-travel protection. Provisions shall be made to prevent over-travel at the highest rise which would endanger the operator or damage the wheel lift truck components.

3.4.6.2 Warning plates (hydraulic unit). When a hydraulic unit is provided, warning plates shall be located near the filler plug of the hydraulic fluid reservoir. The plates shall be marked as specified in 3.8. The manufacturer shall choose hydraulic fluid conforming to either MIL-F-17111 or MIL-H-6083 to fill the reservoir. Hydraulic fluids conforming to MIL-F-17111 and MIL-H-6083 shall not be used together as they are incompatible. If the reservoir is filled with MIL-F-17111 fluid, the plate shall be marked as follows: CAUTION: USE ONLY HYDRAULIC FLUID CONFORMING TO MIL-F-17111. If the reservoir is filled with MIL-H-6083 fluid, the plate shall be marked as follows: CAUTION: USE ONLY HYDRAULIC FLUID CONFORMING TO MIL-H-6083.

3.4.7 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model.

3.4.8 Protective finishes. Exposed metal and working surfaces, except when parts are inherently corrosion resistant, shall be finished to resist corrosion. Exposed metal is defined as metal exposed to atmospheric conditions.

3.5 Performance and product characteristics.

3.5.1 Function. The wheel lift truck shall include a lift mechanism for effective wheel assembly removal and wheel assembly installation, of single and dual wheel and tire assemblies on a raised vehicle. The wheel lift truck shall raise and lower the vertically positioned tire and wheel assembly in relation to floor level. Means shall be provided to level and align the axial axis of the vertically positioned wheel assembly with the axial axis of the vehicle axle or spindle assembly when loaded with dual tires and wheels at safety proof load 1875 lbs. To obtain the specified weight of 1875 lbs, weights may be added to the dual tire and wheel assembly as specified in 4.6.2. The arial axis of the tire and wheel assembly as positioned on a horizontal plane parallel to the plane of the floor shall be able to be inclined up two degrees and down two degrees from the horizontal plane. The level align and the inclination requirements are required to prevent damage to bearings and grease seals and to allow easy removal and installation of the wheel assembly to the vehicle axle or spindle assembly. A means shall be provided to safety and easily maneuver the wheel lift truck while transporting the specified tire and wheel assemblies. Functional components of the wheel lift truck shall be capable of being manually operated and controlled by one person. The operation of the wheel lift truck shall not result in damage to the tire and wheel assembly and the vehicle axle assembly. The dual tire/wheel assembly shall include: hub, brake drum or disc, bearings, cups, races, dual wheels, rims, tires and tubes (if applicable) and all necessary bolts, nuts, retainers, and fasteners. The wheel lift truck shall fully raise and lower smoothly, without any interruptions or jerkiness, under fully loaded and unloaded conditions.

3.5.2 Capacity. The wheel lift truck shall have a load carrying and lifting capacity of not less than 1500 lbs. The wheel lift truck shall be capable of removing from, and replacing on, automotive vehicle axle or spindle assemblies

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single and dual tire and wheel assemblies in a tire size range from 7.00 x 16 thru and including 14.00 x 24. For purposes herein, a tire and wheel assembly may include the tire, tube, rim, wheel, brake drum, and wheel hub assembly.

3.5.2.1 Traversing capacity. The wheel lift truck, when loaded as specified in 3.5.1 to its safety proof load (1875 lbs), shall be capable of being moved at from 15 to 30 feet per minute (f.p.m.) across a one inch high, 25 degree slope rise in the floor and a one inch 90 degree depressed step in the floor at an approach angle which will bring only one caster at a time in contact with the rise and depression. The wheel lift truck shall transverse the rise and depression without any evidence of instability, damage, deformation, breakage, or binding of the lifting mechanism or leakage of the hydraulic fluids.

3.5.2.2 Safety proof load. The wheel lift truck shall be capable of raising, lowering, sustaining and traversing, throughout the specified range of operations, a safety proof load of 1875 lbs as specified in 3.5.1 and 3.5.2, without damage, leakage, or permanent deformation with the exception of the wheel lifting supports as specified in 3.6.4.

3.5.3 Operating temperature. The wheel lift truck and all components thereof shall be able to operate, without damage, within the performance requirements set forth in 3.5 while being subjected to an operating temperature of minus 15 degrees Fahrenheit (F.) and plus 150 degrees F.

3.6 Detail of components. The wheel lift truck shall consist of, but not be limited to, the following component parts.

3.6.1 Frame. The frame shall be made of a metal capable of withstanding without damage the load capacity and performance requirements as specified in 3.5. The frame shall be of such design as to hold the specified wheel and tire assemblies in a stable upright position. The frame shall incorporate handles, located not less than 29 inches above floor level, to permit the operator to pull and push the wheel lift truck when removing, replacing, and conveying wheel and tire assemblies.

3.6.2 Casters. The wheel lift truck shall be provided with either 3 or 4 load carrying casters. The two front (end opposite operator) casters shall be the rigid type. If 4 load carrying casters are provided, the two back casters (operator end) shall be the swivel type. If three load carrying casters are provided, the rear frame structure shall be equipped with one load carrying swivel caster in the center and two smiler outrigger, stabilizing casters fastened at the rear frame structure corners. The load carrying casters shall be at least four inches minimal diameter and shall provide for free movement of the wheel lift truck in any direction over the floor surface under the load conditions specified herein. The wheel lift truck shall be provided with antifriction bearings in the caster swivels and wheels. Caster positioning shall effectively support movement of the wheel lift truck and prevent over-balancing when the load is concentrated near the end of the lifting arms.

3.6.3 Lifting arms. The wheel lift truck shall be equipped with not less than two lifting arms, each with a minimum dual wheel and tire supporting length of 30 inches, exclusive of the frame or supporting members to the end of the lifting arms. Each arm shall provide at least 30 inches that are available for tire tread surface contact support. The lifting arms and all components thereof shall be capable of withstanding without damage the specified load capacity and be of a design to comply with the performance requirements as specified in 3.5. The lift arms shall be capable of lowering the 14.00 x 24 and 7.00 x 16 dual wheel and tire assemblies to

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floor level (Floor level is defined as that plane upon which the casters rest.) The vertical distance, through which the lifting arms shall move the load from the floor level, shall not be less than 4.25 inches. The lifting arms shall be designed to facilitate freely rotating and positioning a wheel assembly prior to and during assembly to vehicle. Means shall also be provided to rotate individual and dual wheel assembly for correct alignment to splines and mounting bolts of vehicle axle. The lifting arms shall incorporate needle or roller bearings to facilitate easy rotation of each wheel of a dual wheel assembly as well as the entire dual wheel assembly.

3.6.4 Lifting mechanism. The wheel lift truck shall incorporate a manually operated mechanical or hydraulic lifting device, located at the rear of the lift, and shall be actuated by a lever or handwheel which shall be accessible to the operator. The actuating device shall be so located, and its operation shall be such, that the resulting action of the lift is not hazardous to the operator. The lifting mechanism shall possess a mechanical advantage which requires a physical force of not more than 50 lbs to raise and lower the lifting arms in the manner specified in 3.5.1, while loaded to the capacity as specified in 3.5.2. Unless otherwise specified (see 6.2.1), the lifting mechanism shall be either a mechanical or hydraulic operated unit at manufacturer's option. The lifting mechanism shall be capable of raising, lowering, and retaining a 1875 pound bad as specified in 3.5.1 at any height within the range specified herein. When loaded as specified in 3.5.1, to a capacity of 1875 lbs, the wheel lift truck shall be capable of withstanding 10 complete lifting and lowering cycles without damage or without permanent deformation with the exception of the wheel lifting supports which shall not deform permanently more than 0.03 inch in any direction at the support extremities.

3.6.4.1 Hydraulic lifting device. If a hydraulic mechanism is provided, it shall be capable of maintaining the load capacity (1875 lbs) specified in 3.5.1 and 3.5.2 at heights within the specified range for a period of not less than 30 minutes without leakage or a change in the height position in excess of 0.125 inch. The height position change shall not exceed 0.021 inch for any five-minute interval. The hydraulic unit shall consist essentially of a power ram, pump reservoir, and valves. The reservoir shall be fitted with a filler plug or plugs.

3.6.4.1.1 Hydraulic fluid. Only hydraulic fluid conforming to MIL-F-17111 or MIL-H-6083 shall be provided in each jack. The seals of the jack's hydraulic unit shall be compatible with the hydraulic fluid chosen and used in the jack.

3.6.4.1.2 Valves. The hydraulic unit shall incorporate a control (release) valve. The release valve shall permit lowering and stopping of the lifting arms at any position along the descent and be controllable to a rate of descent of one inch per minute. The release valve shall be mounted at a protected location or provided with a locking device to prevent accidental lowering of the lifting arms. The release valve shall be accessible and so located that the lowering of the lift shall not be hazardous to the operator.

3.6.5 Wheel retaining device. The wheel lift truck shall incorporate a wheel retaining device. The wheel retaining device shall be capable of securely holding the wheel and tire assemblies in an upright position and prevent the wheel and tire assembly from falling off the wheel lift truck during its functions as specified in 3.5.

3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing of the wheel lift truck shall be in accordance with the contractor's commercial practice provided the following minimum criteria are met or exceeded: all surfaces

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to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, equivalent to the best commercial practice in the vehicle lift industry.

3.8 Nameplate. The nameplate and information plate(s) can be incorporated into one or more corrosion-resistant metallic plate(s) at the machine manufacturer's option in accordance with MIL-STD-130. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The nameplate shall be secured to the machine with permanent fasteners such as screws, bolts, and rivets. The nameplate shall be secured with permanent fasteners at each of the four corners. The nameplate shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number
Load capacity

3.9 Workmanship. Standards of workmanship shall be such that equipment specified herein shall comply with the requirements in Section 3. All components, subassemblies, parts, and the final assembly shall comply with tests in Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operation, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective materials, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified in the contract, first article inspection shall be applied to the preproduction model or initial production item specified (see 3.1). Unless otherwise specified (see 6.2.1), first article

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inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for' acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

a. Product examination (see 4.5).
b. Operational test (see 4.6.1).
c. Lifting mechanism test (see 4.6.7).
d. Function test (see 4.6.2).
e. Packaging inspection (see 4.7).

4.5 Product examination. Visually, dimensionally, and manually examine each item to determine conformance with the requirements of 3.2 through 3.4.8, 3.6 through 3.9. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to identified specified standards, adequacy of markings, proper cleaning, and freedom from defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the item and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.4.7), shall be cause for rejection.

4.6. Test methods.

4.6.1 Operational test. The wheel lift truck shall be loaded as specified in 3.5.1 with 14.00 x 24 or larger dual tire and wheel assembly to at least 1875 lbs and actuated through the entire lifting range. If a hydraulic lifting mechanism is provided, pumping action shall be continued until the over-travel protection device (see 3.4.6.1) has actuated. The control (release) valve (see 3.6.4.1.2) shall be operated to lower the lifting arms, stopping at mid-range, then to floor level. During this test, all components shall function as specified and all components shall show no evidence of damage or permanent deformation, binding of the lifting mechanism, or leakage of hydraulic-type lifting mechanisms.

4.6.2 Function test. For the function test, either two military vehicles or a vehicle axle or spindle shall be used to test the following minimum wheel and tire configurations:

a. Fourteen x 24 or larger dual tire and wheel assembly, as specified in 3.5.1 to facilitate testing of the wheel lift truck for maximum size as specified in 3.5.2, weights shall be added (bolted) to a 14.00 x 24 or larger dual tire assembly or the inner tube or tire shall be filled with liquid to achieve the required test weight of 1875 lbs as specified in 3.5.2.2.

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b. 7.00 X 16 single or dual tire and wheel assembly.

c. The above tire/wheel assemblies shall include hub, brakedrum, or disc bearings, cups, races, dual wheels, rims, tires and tubes (if applicable) and all necessary bolts, nuts, retainers, and fasteners.

The wheel lift truck shall be used to remove, replace, and rotate the wheel and tire assemblies in accordance with the manufacturer's instructions without endangering the operator or damaging the tire, wheel assembly, or axle assembly. The wheel lift truck shall be used to lift from both sides (inboard and outboard of the demounted dual wheel assembly). Function components of the wheel lift truck shall be capable of being manually operated and controlled by one person. During the function testing, the 14.00 x 24 or larger dual tire and wheel assembly shall be held securely using the wheel retaining device per 3.6.5. After removal, while the dual 14.00 x 24 or larger wheel and tire assembly shall be held securely using the wheel retaining device as specified in 3.6.5. After removal, while the dual 14.00 by 24 or larger wheel and tire assembly is on the lift truck, it shall be lowered to the floor (to verify 3.6.3 requirement), then raised at least 4.25 inches from floor level (without external adjustment), and then conveyed for at least 50 feet on a hard level surface floor without tipping or overturning. Then, the wheel lift truck shall be positioned and removed without interference when the wheel assembly is resting on the floor, after which the tire and wheel assembly shall again be placed on the lift truck. The dual tire and wheel assembly shall be rotated in accordance with requirements as specified in 3.6.3. The dual tire and wheel assembly shall then be installed on the vehicle axle spindle. At the conclusion of this test, the wheel lift truck shall show no evidence of damage or be permanently deformed in any way. The wheel lifting supports shall not have deformed permanently more than 0.03 inch in any direction at the support extremities in accordance with 3.6.4.

4.5.3 Traversing test. The wheel lift truck shall be loaded as specified in 3.5.1 with 14.00 x 24 or larger dual tire and wheel assembly to at least 1875 lbs and secured as specified in 3.6.5. The wheel lift truck shall be moved at from 15 to 30 f.p.m. across a one-inch high 15 degree slope rise and a one-inch 90 degree depressed step in the floor. All casters shall traverse the rise and depression at an approach angle which shall bring only one caster at a time in contact with the rise and depression. At the conclusion of the test, the wheel lift truck shall be inspected for damage to determine compliance with 3.5.2.1.

4.6.4 Height sustaining test (for wheel lift truck with hydraulic mechanism only). The hydraulic wheel lift truck shall be loaded as specified in 3.5.1 to at least 1875 lbs and secured in the same manner as specified in 3.5.5. The lifting arms shall be raised to any point within the upper 1/3 position of the working height range and be permitted to remain there for a period of 30 minutes. The height setting shall be recorded at the beginning of the period and every 5 minutes throughout the 30 minute period. On conclusion of the time period, the height setting shall be measured to determine compliance with the requirement of 3.6.4 1.

4.6.5 Release valve control test (for lifting with hydraulic mechanism only). The hydraulic wheel lift truck, in the level position, shall be loaded as specified in 3.5.1 with 14.00 x 24 or larger dual tire and wheel assembly to at least 1875 lbs and secured in the same manner as specified in 3.6.5 and then raised to the maximum working height setting. The release valve shall be adjusted to permit the lowering of the lifting arms. The rate of descent of the lifting arms shall be measured by any applicable method to determine compliance with 3.6.4.1.2. The measuring method shall have an accuracy of within 0.5 percent.

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4.6.6 Operating temperature tests (applies to both hydraulic and mechanical lift trucks.

4.6.6.1 Low temperature operating test. The wheel lift truck shall be placed in an ambient air temperature of minus 15 degrees F. and allowed to stabilize. While at a minus 15 degrees F., the wheel lift truck shall be tested for operational requirements as specified in 4.6.1. During and upon conclusion of the test, the wheel lift truck shall be inspected for evidence of' any damage or permanent deformation, binding of the lifting mechanism, or leakage of hydraulic-type lifting mechanisms (see 3.5.3).

4.6.6.2 High temperature operating test. The wheel lift truck shall be placed in an ambient air temperature of 150 degrees F. and allowed to stabilize. While at 150 degrees F., the wheel lift truck shall be tested for operational requirements as specified in 4.6.1. During and upon conclusion of the test, the wheel lift truck shall be inspected for evidence of any damage or permanent deformation, binding of the lifting mechanism, or leakage of hydraulic type lifting mechanisms (see 3.5.3).

4.6.7 Lifting mechanism test. The wheel lift truck shall be loaded as specified in 3.5.1 with 14.00 x 24 or larger dual tire and wheel assembly to at least 1875 lbs and secured in the same manner specified in 3.6.5. The lifting arms shall be operated for a total of 10 lifting cycles. Each cycle shall consist of raising the lead to the maximum height permitted by the lifting arms and then lowering. On completion of the tests, the wheel lift truck shall be inspected for compliance with 3.6.4.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging of the item to determine compliance with Section 5 requirements.

5. PACKAGING

5.1 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage, or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.2 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the item shall be marked by stenciling, on the part, or by use of tags. Tags shall be waterproofed and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

5.3 Preservation. Preservation shall be Level A or C as specified in the contract (see 6.2.1).

5.3.1 Level A.

5.3.1.1 Cleaning. The wheel lift truck shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.3. 1.2 Drying. The wheel lift trucks shall be dried in accordance with procedures D-1 and D-4 of MIL-P-116.

5.3.1.3 Preservative application. After the wheel lift trucks have been cleaned and dried, preserve all exposed, unpainted, functional surfaces with an application of type P-11 preservative. Exposed unpainted, nonfunctional surfaces shall be preserved with an application of type P-1 or type P-19 preservative. Caster bearings shall be preserved with type P-n preservative and shall be wrapped with barrier material conforming to MIL-B-121, grade A. The barrier material shall be secured with tape conforming to PPP-T-60, type optional.

5.3.1.4 Unit packaging. The wheel lift trucks shall be packaged in accordance with Method I of MIL-P-116. The ram and piston assembly of the hydraulic unit shall be retracted into the cylinder and secured in a closed position. Place accessory tools, lift plate adapters, and tie down equipment in one or more fiberboard boxes conforming to PPP-B-636, class weather-resistant. Adapter hardware shall receive a double wrap of MIL-B-121 barrier material and be packaged with accessory tools. A positive means shall be applied (caps, plugs, tape, etc.) to assure that no fluids will lead out during shipment or storage regardless of the physical position of the lift. Technical data shall be packaged in accordance with Method 1C-1 of MIL-P-116 and attached to the basic unit.

5.3.2 Level C. The wheel lift trucks shall be unit packaged in accordance with MIL-STD-1190.

5.4 Packing. Packing shall be Level A, B, or C as specified in the contract (see 6.2.1).

5.4.1 Level A. The wheel lift trucks shall be individually packed in nailed wood boxes conforming to PPP-B-601, overseas type, or PPP-B-621, Class 2. The complete wheel lift truck shall be cushioned, blocked, and braced in accordance with MIL-STD-1186. Closure, strapping, and skidding shall be in accordance with the appendix of the container specification.

5.4.2 Level B. The packing requirements shall be the same as Level A (see 5.4.1) except that the shipping containers shall be of the domestic type.

5.4.3 Level C. Level C packing shall be in acordance with MIL-STD-1190.

6. NOTES

6.1 Intended use. The wheel lift truck covered by this specification is manually operated and is intended for use in removing, replacing, and conveying automotive-type wheel and tire assemblies.

6.2 Ordering data.

6.2.1 Acquisition data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2).
c. First article, when required (see 3.1).
d. Special safety devices, when required (see 3.4.6 and 5.5).
e. When mechanical or hydraulic unit only is acceptable (see 3.6.4).

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h. First article, if different (see 4.3).
i. Quality conformance inspection, if different (see 4.4).
j. Level of preservation and packing, if different (see 5.3 and 5.4).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1654, should be specified on a DD Form 1423 incorporated in the contract.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provision of the Federal Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

6.5 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other safety and health requirements, should be specified by the user.

6.6 Sampling. Sampling for quality conformance inspection should be performed as follows. Lots should be accepted with zero defects and rejected if one or more defects are found.

Inspection or Test	Lot or Batch Size	Number randomly selected for inspection
Product examination (4.5)	1-150	100 percent inspection
Operational test (4.6)	1-150	8
Packaging inspection (4.7)	1-150	5

The lots should not exceed the maximum sizes indicated above.

6.7 Subject term (key word) listing.

Mechanical lifting
Hydraulic lifting
Over travel

5.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodian:
Army - AL
Navy - YD
Air Force - 99

Preparing activity:
Army - AL

Review activities:
Army - SM

Project No. 4910-0645

User activities:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-T-14537G, Truck, Lift, Wheel

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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